Enabling, Coordinating, Collaborating: A Three-Layer Framework for Disability-Centered Human-Agent Collaboration

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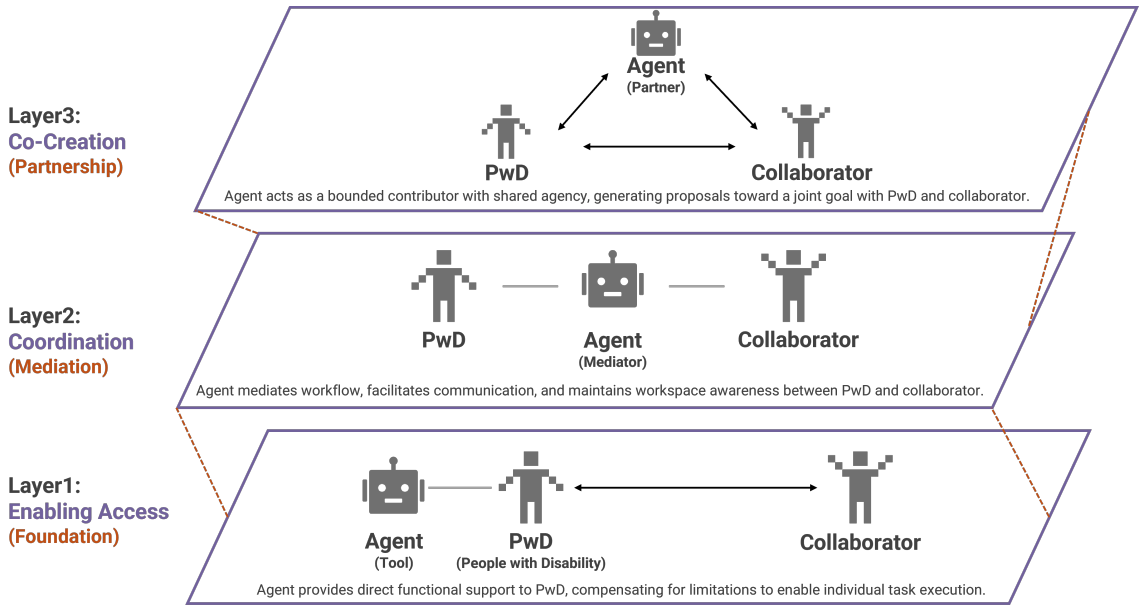

Fig. 1. Disability-Centered Human-Agent Collaboration Framework

AI accessibility tools have mostly been designed for individual use, helping one person overcome a specific functional barrier. But for many disabled people, complex tasks are accomplished through collaboration with others who bring complementary abilities, not solitary effort. This collaborative reality has received limited attention in accessibility research or human-agent collaboration. We propose a three-layer framework that rethinks AI's role in ability-diverse collaboration: as an information access provider that establishes shared ground across modalities, a coordination mediator that supports workflows between collaborators with different abilities, and a co-creative partner that contributes its own capabilities toward shared goals. Grounded in the Ability-Diverse Collaboration framework and empirical research with disabled creators, this framework extends the "agents as remote collaborators" vision by centering the collaborative, interdependent ways disabled people already work.

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CCS Concepts: • **Human-centered computing → Accessibility theory, concepts and paradigms**.

Additional Key Words and Phrases: human-agent collaboration, disability, CSCW, accessibility, co-agency, AI coordination

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1 Introduction

Large language model (LLM) agents are increasingly integrated in everyday workflows. Across writing, coding, and creative production, people use these systems not only to generate outputs but to plan, iterate, and coordinate complex tasks. This shift has prompted a broader reframing of AI from tool to collaborator in HCI and CSCW, where the quality of interaction depends on shared context, clarity of role, and coordination over time [2, 14, 17, 29]. People with disabilities are active participants in this transition. Blind users regularly employ systems such as Be My AI and ChatGPT for information access and day-to-day decision support [1, 23]. Video creators with sensory impairments use LLM-based tools for scripting, captioning, and visual description across production workflows [27], suggesting that AI is becoming part of end-to-end creative practice rather than isolated assistive use. However, AI accessibility tools and much of the surrounding HCI research remain focused on individual assistance: compensating for a specific functional limitation through a one-person, one-tool interaction model. This framing has produced important technical advances, but it misses how access is actually achieved in social settings. Disability scholarship has long pushed back against this individualist view. Bennett et al. [6] argue that interdependence, not independence alone, better captures how disabled people actually navigate daily life, and should sit at the centre of assistive technology research. Xiao et al. map recurring patterns in ability-diverse collaboration, showing that collaboration is itself the mechanism through which access gets produced [26]. Empirical work tells a similar story. Blind runners coordinate with sighted guides through continuous, fine-grained communication that weaves together voice cues and bodily movement [5]. Creators with sensory impairments often split the work of video production across multiple stages with trusted partners, from filming through editing and publishing [27]. These are not edge cases. They show how, for many disabled people, collaboration is the ordinary infrastructure behind complex professional and creative work.

In this position paper, we propose a three-layer framework that addresses this gap. Layer 1, Information Access, concerns modality-adapted, equivalent access to task-relevant information. Layer 2, Coordination, concerns how AI can mediate workflows, communication, and handoffs among collaborators with different abilities. Layer 3, Co-Creation, moves beyond support to consider AI as a bounded contributor that advances shared goals alongside human partners. Together, these layers extend both accessibility and human-agent collaboration research by centring interdependence in disabled people's everyday practices and by treating triadic collaboration as an explicit design target.

2 Background and Positioning

2.1 Ability-Diverse Collaboration

The Ability-Diverse Collaboration (ADC) framework [26], based on a systematic review of 117 HCI papers, distinguishes two collaboration modes: ability sharing, where an ability provider contributes a required ability (e.g., sight for visual editing) while the borrower contributes directive input and evaluative judgment; and ability combining, where all collaborators serve as ability providers toward a shared goal. The framework also identifies four technology categories:

ability channels that transmit abilities, ability supporters that augment abilities, communication supporters that facilitate coordination, and ability combiners that integrate diverse abilities into shared outputs. These categories provide a vocabulary for mapping AI's potential roles in collaborative settings. An ability-based design perspective [25] complements this vocabulary by shifting focus from what users cannot do to what they can, a principle that extends naturally to collaborative configurations.

2.2 Human-Agent Collaboration and Its Limits

The vision of agents as collaborative partners builds on grounding theory [9] and workspace awareness [13, 15], two established frameworks for how collaborators establish mutual understanding and maintain awareness of each other's activities. Early work on mixed-initiative interaction [17] and the challenges of remote collaboration [21] set the stage for recent extensions that treat human-agent collaboration as structurally similar to distributed human teamwork [2, 29]. However, these foundations carry assumptions that break down in ability-diverse settings. Grounding theory presumes shared perceptual access to the same evidence, which does not hold when one collaborator is blind and another sighted. Workspace awareness presumes a common modality for presenting awareness information. And the "remote collaborator" metaphor models a dyad, whereas in ability-diverse settings, AI typically enters an existing partnership, creating a triad with fundamentally different coordination demands. These gaps motivate our framework.

3 A Three-Layer Framework

We propose three layers of AI involvement in ability-diverse collaboration, each progressively deeper (Figure 1). The layers are nested: coordination (Layer 2) depends on information access (Layer 1), and co-creation (Layer 3) depends on both. We ground each layer in the ADC technology categories and illustrate with examples from multiple domains.

3.1 Layer 1: Information Access

In ability-diverse collaboration, information asymmetry is often the root barrier. A blind video creator cannot see the timeline; a deaf collaborator in a design team cannot hear verbal feedback during a critique session. This asymmetry forces delegation: the person must rely on a partner who becomes the sole gatekeeper of critical information, constraining their agency to what they are told rather than what they can independently access. Research on accessible collaborative writing [11] and collaboration awareness in document editing [18] has begun to address fragments of this problem, but typically for a single user rather than a collaborative pair. At this layer, AI functions as an ability supporter, providing equivalent, modality-adapted information to all collaborators. This does not mean identical information, but functionally equivalent access sufficient to support informed decision-making. AI-generated video descriptions can give a blind creator scene composition and colour palette information; real-time captioning can give a deaf team member access to rapid-fire design discussion. The key shift is from delegation to informed direction. When a blind creator has independent access to visual information, the collaboration dynamic changes: rather than asking "what is on screen?", they can say "let us adjust the framing here." In CSCW terms, this layer establishes common ground [9] across modalities—the shared informational foundation without which meaningful collaboration cannot occur.

3.2 Layer 2: Coordination

Even when all collaborators can access the same content, ability-diverse collaboration requires substantial coordination work: negotiating roles, managing handoffs, tracking each other's activities, and resolving differences in judgment.

This coordination work is often invisible and burdensome, what Branham and Kane[8] call the "invisible work of accessibility" in mixed-ability workplaces.

At this layer, AI functions as a communication supporter, mediating workflow between collaborators with different abilities. Concretely, this means: maintaining task state across modality boundaries (converting a visual editing history into an audio summary for a blind creator), providing awareness cues adapted to each collaborator's sensory modality, and supporting structured handoffs when collaborators share a single device—a common setup in low-resource contexts [4, 27]. Prior work on paired remote sighted assistance [28] shows that even human-human coordination in these settings is effortful; AI-mediated coordination could reduce this burden.

This layer addresses workspace awareness [15] in ability-diverse settings. A sighted collaborator can glance at a timeline to see where edits were made; a blind collaborator needs this translated into audio or haptic form. AI serves as a persistent translator of workspace state across modalities, so all collaborators maintain a coherent picture of the process. The shift is from ad hoc verbal negotiation to structured, AI-mediated workflow continuity, reducing the invisible labour that falls disproportionately on disabled collaborators and their partners.

3.3 Layer 3: Co-Creation

In Layers 1 and 2, the collaboration model remains ability sharing: the creator holds the primary goal, and both the human collaborator and AI serve supporting roles. Layer 3 marks a shift toward ability combining, in which AI contributes as a third participant with distinct capabilities directed toward a shared goal.

At this layer, AI generates, suggests, and contributes. In video production, this might mean generating B-roll options that match a creator's described aesthetic for the team to evaluate together, or producing rough-cut assemblies based on the creator's audio annotations that the sighted collaborator then refines visually. In collaborative music-making between musicians with different motor abilities, AI could generate rhythmic accompaniment patterns that complement each human performer's physical playing style. Emerging work on human-AI collaboration in remote sighted assistance [19, 30] explores related configurations, though not yet in co-creative contexts.

The critical design requirement is that AI contributions are proposals, not decisions: transparent, modifiable, and clearly distinguished from human-authored work. Research with disabled creators consistently shows that they maintain creative control even in heavily collaborative workflows [27]. When AI enters as a co-creative partner, it must preserve this agency. Extending the ADC notion of "ability provider" to a non-human agent is a deliberate theoretical move: AI does not share abilities in the same relational, embodied sense that human collaborators do [7]. But the functional role—contributing a required capability toward a shared goal—maps onto the ability combining model, and naming it as such allows us to use the ADC vocabulary to reason about triadic configurations that current CSCW frameworks do not address.

4 Research Agenda

The framework opens questions at the intersection of accessibility and human-agent collaboration. We highlight four that we believe are especially pressing.

What counts as equivalent information across modalities? A text description of a video frame conveys different things than seeing it. How should AI calibrate detail level, and who controls that calibration? How do cultural and linguistic contexts shape what counts as adequate access?

How does reducing information asymmetry reshape collaboration? If Layer 1 gives all collaborators equivalent access, do established roles shift? Does a blind creator who can now independently access visual information take on new tasks, or do existing divisions of labour persist? What new tensions emerge?

How is agency negotiated in triadic configurations? When does AI shift from supporting collaboration to transforming it? How do collaborators negotiate creative ownership when AI has generated content that both humans refine? This question is especially sharp for disabled creators who have fought to maintain their creative authority. Research on algorithm aversion [12] and complementary team performance [3] suggests that trust calibration in human-AI teams is already complex; ability-diverse settings add further dimensions.

What extensions to CSCW theory are needed? The "remote collaborator" metaphor is productive, but a collaborator that perceives through multiple modalities simultaneously, never tires, and has no lived experience of disability is a fundamentally different kind of partner. How do common ground and workspace awareness need rethinking for triadic, ability-diverse settings [9, 15], and what new concepts might be required? Parasuraman et al.'s taxonomy [22] of automation levels may offer a complementary lens for specifying AI's degree of involvement across the three layers.

5 Discussion

The framework makes two claims. First, accessibility should be understood as collaborative infrastructure, not only individual assistive capability. When access is generated through collaboration, as empirical research consistently demonstrates [6, 24, 26], AI systems must be deisgned to facilitate that collaboration rather than merely to address individual deficiencies. This new way of looking at things fits with rights-based approaches to disability that put more emphasis on participation than on fixing problems. Second, the maturity of AI in disability contexts should be evaluated through socio-technical outcomes: role transparency, repairability of breakdowns, and fair distribution of coordination labour, not only individual assistive task performance. These claims carry practical implications. System designers should make collaboration states: roles, intentions, uncertainties, handoff statuses, and escalation pathways. Based on current tools that deal with parts of this problem [10, 11, 18], we say that future systems should show collaboration metadata: who contributed what, what the AI isn't sure about, and where human judgement is needed in ways that are appropriate for each participant. Another implication has to do with governance. If triadic co-agency is a goal for design, organisations need policies and interfaces that make it easy to negotiate and contest. Without these, advanced AI may replicate paternalistic patterns using a novel lexicon—a concern frequently articulated by disability scholars regarding assistive technologies [16, 20]. We note several limitations. It is more of a concept than a fact; the layers need to be tested through design research and user studies with disabled people who work with them. The framework has not yet addressed how its layers interact with power asymmetries beyond ability differences (e.g., employer-employee, expert-novice). Lastly, our examples mostly come from sensory impairments and creative production. It's still important to expand the framework to include cognitive, motor, and other disability experiences.

6 Conclusion

Human-agent collaboration research has made real progress on trust, coordination, and shared agency, but has largely overlooked a basic fact: for many disabled people, collaboration is not occasional but structural, the ordinary infrastructure through which complex work gets done. Accessibility research understands this deeply but has not yet engaged with AI agents as collaborative partners. The question that connects both fields is deceptively simple: what happens when an AI agent joins a collaboration already shaped by difference, interdependence, and hard-won creative agency?

Our three-layer framework proposes that AI's involvement is a progression, not a single relationship. At the foundation, AI establishes shared informational ground across modalities. In the middle layer, it mediates handoffs and reduces the invisible coordination labour that ability-diverse partnerships absorb. At the deepest level, it contributes as a bounded creative participant whose proposals human collaborators evaluate and refine on their own terms. The framework is conceptual and needs empirical grounding—through design research with disabled collaborators, engagement with power asymmetries beyond ability differences, and extension beyond sensory impairments and creative production. But we believe its core move like centring interdependence rather than individual deficit, and treating triadic ability-diverse collaboration as an explicit design target, will opens productive ground for both fields to build on.

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